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Slow drift rate predicts ADHD symptomology over and above executive dysfunction

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ABSTRACT

Slow drift rate has become one of the most salient cognitive deficits among children with ADHD, and has repeatedly been found to explain slow, variable, and error-prone performance on tasks of executive functioning (EF). The present study applies the diffusion model to determine whether slow drift rate better predicts parent and teacher ratings of ADHD than standard EF metrics. 201 children aged 8–12 completed two tests of speeded decision-making analyzed with the diffusion model and two traditionally scored tests of EF. Latent EF and drift rate factors each independently predicted the general ADHD factor in a bifactor model of ADHD, with poor EF and slow drift rate associated with greater ADHD symptomology. When both EF and drift rate were entered into the model, slow drift rate (but not EF) continued to predict elevated symptomology. These findings suggest that using drift rate to index task performance improves upon conventional approaches to measuring and conceptualizing cognitive dysfunction in ADHD. Implications for future cognitive research in ADHD are discussed.

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ADHD; drift rate; diffusion model; executive function; structural equation modeling

Executive dysfunction among children with Attention-Deficit/Hyperactivity Disorder (ADHD) is among the most extensively documented findings in clinical research. Nevertheless, meta-analytic effect sizes of executive functions (EF) deficits are moderate at best (Frazier et al., 2004; Martinussen et al., 2005; Willcutt et al., 2005), and only 30–50% of children with ADHD underperform on tasks of EF (Nigg et al., 2005). Such low base rates are counterintuitive considering that a diagnosis of ADHD requires daily, real-world, and cross-situational impairment in actions that are presumed to require EF (e.g., following multi-step instructions, switching between tasks, allowing others to finish speaking). Indeed, EF task performance is only weakly correlated with parent and teacher reports of ADHD symptomology (Gerst et al., 2017; Jonsdottir et al., 2006). Although this may to some extent reflect both method variance and fundamental differences in the constructs being measured (e.g., inhibitory control vs. behavioral inhibition; Cyders & Coskunpinar, 2011; Cyders & Smith, 2007; Sharma et al., 2014), these constructs are unlikely to be entirely independent.

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One reason for the modest effect sizes of EF impairment and the discrepancies with behavioral symptomology may be the manner in which executive dysfunction has historically been captured. Namely, slow mean response times (MRT) or low accuracy. For example, working memory (WM) is commonly indexed by accurate item recall, while set shifting is measured by switch cost – the additional amount of time required to correctly respond on trials in which the response “rule” changes from the previous trial. This selectivity in metric of choice ignores the fact that RT and accuracy are non-independent descriptors of the same behavioral response which, when examined separately, can result in qualitatively different interpretations of cognitive abilities. High accuracy rates often come at the cost of slow RTs, and conversely, fast RTs are often accompanied by high error rates. Considered independently, slow RT could be misconstrued as a cognitive deficiency rather than a byproduct of differences in task approach (i.e., greater caution in responding). Conversely, equivalent mean RTs between two groups could be taken to suggest comparable cognitive proficiency if error rates are ignored. The inability to integrate speed and accuracy into a set of unitary indices of performance essentially prevents a full and accurate description of performance.

Furthermore, the common use of mean or median RT as an index of speed on EF tasks is in and of itself problematic because RT distributions are positively skewed. Among children with ADHD, they are even more so. Variance in RT (e.g., standard deviation of RT (SDRT), or the coefficient of variation (CV; defined as SDRT/MRT)) is commonly larger among children with ADHD. The same is true for the ex-Gaussian parameter tau, which explicitly represents the size of the tail of the distribution (Epstein et al., 2011; Hervey et al., 2006). However, these metrics are all based solely on RT to correct responses and exclude RT to incorrect responses. Thus, as with MRT, the reliance on only a portion of the data renders them unable to fully detail task performance.

And finally, RTs are influenced by multiple cognitive subprocesses that occur between the appearance of a stimulus and the participant’s response. These subprocesses include the time spent encoding information, gathering evidence to support the decision, preparing and executing a motor response as well as strategic or person-specific emphases on responding quickly or accurately (Boywitt & Rummel, 2012; Dutilh et al., 2013, 2011; Ratcliff et al., 2012, 2001; Schmiedek et al., 2007). Unfortunately, interpretation of single-index performance metrics, including MRT, SDRT, CV, Shifted Wald, and ex-Gaussian parameters, precludes identification of the process-level source of performance differences. In sum, the common use of process-nonspecific metrics, as well as partial data analysis and/or interpretation, invariably produces an incomplete picture of cognitive abilities in ADHD.

Computational models of decision-making such as the diffusion model (DM; Ratcliff, 1978) present an advantageous descriptive alternative because unlike single-metric indices, they consider the entirety of speed and accuracy performance data. The DM parameters are based on the *shapes* of the RT distributions for both correct and error responses, and are therefore also heavily influenced by RT variability (i.e., distributional skew). Conceptually, the DM assumes that following a period of stimulus encoding, binary decisions are made through a noisy process in which evidence supporting one of the two potential response options is continuously accumulated until crossing an “upper” (i.e., accurate) or “lower” (i.e., error) response boundary. Once one of these thresholds is crossed, the corresponding motor response is initiated. The efficiency with which one gathers this decisional evidence is represented by the drift rate (ν) parameter, while boundary separation (a) reflects the distance between the decisional thresholds and the

amount of evidence desired or required to make a decision. Boundary separation is therefore generally interpreted as an index of response caution; the larger the separation, the more evidence is accumulated before initiating a response. Finally, nondcision time (*Ter*) is a composite of time spent on encoding, motor response preparation/execution, and all other processes unrelated to the active decision-making process.

The psychological interpretations of the DM parameters have been extensively validated (Matzke & Wagenmakers, 2009; Ratcliff & McKoon, 2008; Voss et al., 2004; Wagenmakers, 2009) and therefore offer much-needed improvements upon extant practices in cognitive research. For example, common metrics used to measure RT variability (e.g., tau, CV) are ambiguously or inconsistently defined (Matzke & Wagenmakers, 2009), and ultimately provide limited insight into the cognitive source(s) underlying elevations in RT variability in populations such as ADHD. Additionally, the biological plausibility of the DM's conceptualization of decision-making is also advantageous in that it can address a significant limitation of theoretical models of EF: the historical invocation of neurologically implausible "homunculi" that flip the proverbial neural switches to engage task-relevant cognitive operations (Baddeley, 1996; Monsell & Driver, 2000). Thus, a diffusion model framework uses computationally defined parameters to identify the cognitive subprocesses that independently and collectively work together to influence complex behavior.

Over the past several years, research applying the DM to ADHD has documented slower drift rates relative to typically developing individuals (Feldman & Huang-Pollock, 2020; Huang-Pollock et al., 2020; Karalunas & Huang-Pollock, 2013; Karalunas et al., 2012; Metin et al., 2013; Mulder et al., 2010; Salum, Sergeant et al., 2014; Salum, Sonuga-Barke et al., 2014; Weigard & Huang-Pollock, 2014). Although faster *Ter* and inflexible response thresholds have also been found (Feldman & Huang-Pollock, 2020; Karalunas et al., 2014; Mulder et al., 2010; Weigard & Huang-Pollock, 2014), they are reported inconsistently and with far smaller effect sizes. Among the DM parameters, drift rate is also the most consistent predictor of working memory (encompassing functions such as serial reordering, updating, and dual-processing), inhibition, and switching (Schmiedek et al., 2007; Schmitz & Voss, 2012; Schuch, 2016; White et al., 2014; Willcutt et al., 2005), and mediates the association between ADHD and poor EF task performance (Karalunas & Huang-Pollock, 2013; Weigard & Huang-Pollock, 2017). However, it remains unclear whether drift rate's predictive utility exceeds that of the EF construct that has dominated the ADHD literature for decades. Indeed, to facilitate identification of frank brain injury, tasks of EF were originally designed to measure multiple component processes to increase the probability that a breakdown in any stage of the behavioral sequence would be detected (Lezak, 1995). It may be that the very broadness of EF makes it a better marker of dysfunction than narrower constructs like drift rate.

A final consideration is that virtually all of the literature using the DM has treated ADHD categorically (i.e., comparing cases and controls). This contradicts well-replicated evidence from child, adolescent, and adult samples supporting ADHD as a dimensional syndrome optimally measured using a bifactor structure (DuPaul et al., 2016; Goh et al., 2020; Gomez et al., 2018). Most commonly, this structure includes latent inattentive and hyperactive factors (comprised of the nine inattentive and nine hyperactive symptoms, respectively) and a general ADHD factor onto which all 18 symptoms load. One meta-analysis found that this general factor accounted for an average of 72% of the explained

common variance (ECV) in ADHD symptoms across studies (i.e., variance after partialling out effects of the specific factors), while variance captured by the specific factors was low and inconsistent (3–30%) (Arias et al., 2018). The general factor has also been shown to better predict observed inattentive, hyperactive, and total ADHD symptoms ($\beta = .84-.88$) than the specific factors ($\beta = .02-.22$) (Willoughby et al., 2015), supporting it as a strong dimensional representation of real-world behavioral dysfunction in ADHD.

The current study utilizes structural equation modeling (SEM) to determine the extent to which ADHD symptomology as measured by a general ADHD factor can be individually and collectively predicted by drift rate and standard EF metrics. Consistent with extant literature, latent factors for slow drift rate and poor EF would both be expected to individually predict higher ADHD symptomology. However, because drift rate indexes a clearly defined psychological process (i.e., evidence accumulation speed) and is derived using the entirety of available performance data, it may be the better quantitative descriptor of the specific neurocognitive processes driving cognitive and behavioral dysfunction in ADHD. If this is indeed the case, slower drift rates would be expected to continue to predict greater general ADHD symptomology even after EF as indexed by standard methods is entered into the model. Conversely, the direct association between traditional EF metrics and ADHD symptomology would not be expected survive the inclusion of drift rate as an additional independent predictor.

Method

Participants

Two hundred and one 8-12-year-old children ($n = 75$ girls) were recruited from Centre and Dauphin counties in Pennsylvania to participate in a larger study on attention and learning conducted at The Pennsylvania State University. Children were ethnically representative of the region: 72.64% Caucasian, 4.48% Caucasian/Hispanic, 9.45% African American/non-Hispanic, 1.49% African American/Hispanic, 1.00% Other Hispanic, 1.99% Asian, 8.95% mixed/unknown. Exclusion criteria included a current prescription for a non-stimulant psychoactive medication; a parent-reported history of head injuries, psychosis, neurological, developmental, intellectual, or sensorimotor disabilities; and an estimated Full Scale IQ (FSIQ) below 80 based on a two-subtest short form (i.e., vocabulary and matrix reasoning, predictive validity = 0.87) of the Wechsler Intelligence Scale for Children (WISC-IV; Wechsler, 2003). Anxiety, depression, oppositional defiant, and conduct disorders were not exclusionary.

Inclusion in the study required that both parent- and teacher-reported behaviors either: (1) exceeded the 85th percentile (T -score > 60) on the Inattention, Hyperactivity, or ADHD subscales of the Conners' Rating Scales (Conners, 2001) or Behavior Assessment System for Children (BASC; Reynolds & Kamphaus, 2004), or (2) fell below the 80th percentile (T -score ≤ 58) on the same measures.

Procedures

The following tasks were part of a larger neuropsychological testing battery completed by participants over the course of two 3-hour laboratory visits. All children taking stimulant

medications completed a 24 to 48-hour medication-free “washout” prior to testing ($n = 54$; mean = 110.55 hours). Informed written parental consent and child verbal assent were both obtained prior to participation. All data were collected in compliance with human subjects’ approval from the Pennsylvania State University Institutional Review Board (IRB#32, 126).

Measures

ADHD symptomology

Parent and teacher report on the ADHD Rating Scale (ADHD-RS; DuPaul et al., 1998) were used as an index of inattentive and hyperactive/impulsive symptoms of ADHD. For each of the 18 items, parents and teachers rated children on a 4-point-Likert scale (0 = “never/rarely” to 3 = “very often”). Ratings were summed across raters to account for symptom (0 = min, 6 = max) and total (0 = min, 108 = max) severity across contexts.

Measures of executive function

We simultaneously evaluated performance on two tasks assessing working memory and set shifting, two prototypical EFs with meta-analytic effect sizes for ADHD-related deficits among the largest across all neuropsychological domains (Martinussen et al., 2005; Willcutt et al., 2005; Wright et al., 2014). Bifactor models of EF have shown that shared variance across WM and set shifting tasks is captured by a “common EF” factor, which better predicts psychopathology than does variance unique to either construct (Friedman et al., 2018, 2020; Hatoum et al., 2018; Karr et al., 2018; Miyake & Friedman, 2012). Following this work, we combined our tasks into a latent EF factor in order to capture this shared variance and create the most stringent test of our hypothesis of drift rate as a superior predictor of ADHD.

Working memory. The WISC-IV Working Memory Index (WMI), comprised of both the digit span (DS) and letter-number sequencing (LNS) subtests, was used to index WM. Trained research assistants recited strings of digits (i.e., for DS) or randomly arranged digits and letters (i.e., for LNS) to children at a rate of one per second. Children were then asked to recite digit strings as presented (i.e., DS forward) or in the reverse order of presentation (i.e., DS backward). On letter-number sequencing, children are asked to first recite the numbers in ascending order, followed by the letters in alphabetical order. For all tasks, string length increased from two to nine, and the task was discontinued after two (DS) or three (LNS) consecutive incorrect trials at a given string length.

Set shifting. Set shifting was evaluated using a variant of the NAVON task (Navon, 1977). In this task, figures consisting of a “global” figure (e.g., a square) constructed using smaller “local” figures (e.g., circles) were presented on a computer screen one at a time. Children were instructed to press a key on a response box indicating whether the global (i.e., if the figure was blue) or local (i.e., if the figure was yellow) shape was a circle, square, triangle, or “x”. Participants completed 144 trials, half of which required shifting from examining global to local shapes or vice versa (i.e., a color change from the previous trial). Set shifting was indexed by switch cost, calculated by subtracting the average RT on

trials without a color change (i.e., nonswitch trials) from the average RT on trials with a color change (i.e., switch trials).

Trials faster than 200 ms were discarded as anticipatory. Remaining (accurate) trials were first winsorized to 8,500 ms, and RTs remaining more than three standard deviations (SD) above each child's mean RT were then winsorized to exactly three SDs away (Irwin et al., 2019; Miyake et al., 2000). All trimming affected 3.59% of trials.

Measures of processing efficiency

Processing efficiency, indexed by drift rate, was measured by applying the diffusion model to a perceptual decision task and a mental rotation task previously analyzed using evidence accumulation models (Feldman & Huang-Pollock, 2020; Larsen, 2014; Provost & Heathcote, 2015).

Choice discrimination. Children completed five 40-trial blocks (200 trials total) of a stopping task. On 75% of trials (“go” trials), they were asked to indicate whether an “X” or an “O” had appeared on the screen by pressing the corresponding keys on a response box as quickly and as accurately as possible. An auditory tone accompanying the letter stimuli on the remaining 25% of trials signaled children to withhold a response (“no-go” trials). Stimuli remained on the screen for 3,000 ms or until the child registered a response. Time-outs were removed, as were anticipatory trials faster than 200 ms (0.44% of data). Stopping tasks have been previously hypothesized to consist of two partially independent tasks: (1) the “go” task of discriminating between “X” and “O” and (2) the “stop” task in which children need to refrain from engaging in the “go” decision process (Logan et al., 1997; Schachar et al., 2004; Verbruggen & Logan, 2008). Eliminating stop trials from analysis, as well as go trials immediately preceded by a stop trial, thereby minimizes any potential influence of the stopping condition on the data and ensures the most reliable estimation of diffusion model parameters. This left 107 choice discrimination (“go”) trials available for diffusion modeling.

Mental rotation. Children completed a mental rotation task in which they were asked to make a parity judgment between a standard and simultaneously presented target stimulus via key press on a response box. The standard stimulus was an upright stick figure with a yellow dot on the figure's right hand and a blue dot on the figure's left foot. Targets were either identical or mirror images of the standard stimulus and were rotated to one of eight angular rotations (i.e., 0°, 45°, 90°, 135°, 180°, 225°, 270°, or 315°). Each orientation was presented 18 times (9 mirror, 9 identical) for a total of 144 trials across 3 blocks. Visual feedback (i.e., “Right!” or “Wrong!”) was provided for 1,000 ms between trials. Given that outliers on either end of the distribution can influence parameter estimation (Ratcliff & Tuerlinckx, 2002), and following prior mental rotation studies (Feldman & Huang-Pollock, 2020; Provost et al., 2013), trials faster than 300 ms or slower than 7,000 ms were removed from analyses (6.20% of the data).

Data analytic plan

Children were removed from analyses if they were inadequately engaged in any of the four tasks; considering all tasks were largely straightforward, this was defined as either:

(a) more than 25% of trials (within-task) affected by trimming procedures, or (b) performance falling below the minimum accuracy thresholds of 75% for the choice discrimination task, 65% for the mental rotation task, and 60% for the NAVON task. These thresholds were selected due to clear breaks in the between-subjects distributions and are broadly consistent with those used in previous studies (Herlihey et al., 2013; Verbruggen, Stevens et al., 2014; Vogan et al., 2014). As might be expected, 37 of the 41 children excluded based on task performance met full diagnostic criteria for ADHD. These children were slightly younger than those with ADHD who were included in analyses ($M = 8.70$ vs. 9.67 years, $F(1,179) = 20.26$, $\eta^2 = .10$, $p < .001$), but did not differ in IQ or number of inattentive or hyperactive symptoms (all $p > .29$, all $\eta^2 \leq .006$). Removing these participants resulted in a final N of 201 as previously reported.

Given the sensitivity of SEM to extreme outliers, final performance values deviating more than three SDs from within-task means were winsorized to values exactly three SD away (Miyake et al., 2000; Van Der Ven et al., 2013). This was the case for $<0.01\%$ of the data.

Diffusion model

The Fast-dm modeling program (Voss & Voss, 2007) was used to compute drift rate for the choice discrimination and mental rotation tasks. The program successively compares empirical cumulative distribution functions (CDFs) derived from trial-by-trial RT and accuracy data to ones derived from various estimations of DM parameter values until optimal fit and individual parameter estimates are identified. In line with prior studies applying the diffusion model to these tasks (Feldman & Huang-Pollock, 2020; Karalunas & Huang-Pollock, 2013), only drift rate, boundary separation, and nondecision time were allowed to vary. Inter-trial variability parameters contribute little to the shape of the distribution and were therefore fixed at zero (Voss & Voss, 2007).

Data analytic approach

Modeling was carried out in the lavaan package in R (Rosseel, 2012) using a weighted least squares means and variance adjusted (WLSMV) estimator, as is recommended for use with ordinal data (Brown, 2015). WLSMV has been found to produce reliable parameter and standard error estimates in sample sizes of 200 participants (Flora & Curran, 2004). All subjects included in analyses had complete data. Nonsignificant chi-square, CFI and TLI above .95, RMSEA below .07, and SRMR below .08 were considered indicative of good model fit (Hooper et al., 2008).

To assess the quality of the bifactor model beyond standard fit indices, we also considered: (1) explained common variance (ECV; Brunner et al., 2012), used to index unidimensionality of the general factor; (2) percent of uncontaminated correlations (PUC), used in conjunction with ECV to evaluate potential parameter bias; (3) coefficient omega hierarchical (ω_H ; McDonald, 1999) and coefficient omega subscale (ω_S ; Reise, 2012), indices of factor interpretability for the general and specific factors, respectively; and (4) H-index (Hancock & Mueller, 2001), used to evaluate construct reliability. The general ADHD factor has typically performed well on these indices, while the specific inattentive and hyperactive factors tend to be psychometrically weaker and therefore more difficult to interpret (Arias et al., 2018). In the present study, the following criteria were used to indicate adequate general factor reliability and interpretability as

a representation of ADHD symptomology: $ECV > .60$, $PUC < .80$, $\omega_H > .75$, and $H\text{-index} > .70$ (Canivez, 2016; Reise et al., 2010, 2013; Stucky & Edelen, 2014).

Results

Complete sample descriptives and task performance data are summarized in Table 1. Correlations of manifest variables are provided in Table 2.

Bifactor model of ADHD

A confirmatory factor analysis (CFA) was conducted to examine a bifactor model of ADHD. Summed parent and teacher ratings of the 9 inattentive and 9 hyperactive symptoms on the ADHD-RS loaded onto separate orthogonal latent factors, with all 18 symptoms also loading onto a general ADHD symptomology factor (also orthogonal). Negative residual variance rendered the initial model inadmissible. Examination of

Table 1. Descriptives.

	Mean (SD)	Min/Max	Skew	Kurtosis
Age	9.64 (1.24)			
IQ	103.22 (11.88)			
#Male, Female	126, 75			
#ADHD Subtype (H, I, C) ^a	5, 57, 82			
#ODD ^a , CD ^a	55, 11			
#GAD ^a , MDD ^a , DD ^a	23, 7, 5			
ADHD-RS Inattentive Symptoms (Total)	24.95 (14.36)	0/50	-.32	-1.16
Careless Mistakes	3.25 (1.66)	0/6	-.23	-1.03
Sustains Attention	2.95 (1.89)	0/6	-.24	-1.14
Listens	2.19 (1.64)	0/6	.16	-1.15
Follows Through	2.65 (1.91)	0/6	.06	-1.22
Disorganized	2.81 (1.91)	0/6	-.14	-1.14
Avoids	2.64 (1.83)	0/6	-.05	-1.20
Loses	2.31 (1.76)	0/6	.33	-.77
Distracted	3.56 (1.95)	0/6	-.46	-1.12
Forgetful	2.58 (1.75)	0/6	-.03	-1.03
ADHD-RS Hyperactive Symptoms (Total)	16.46 (13.00)	0/44	.33	-1.13
Fidgets	2.57 (1.99)	0/6	.10	-1.22
Leaves Seat	1.76 (1.76)	0/6	.66	-.64
Runs/Climbs	1.05 (1.26)	0/5	.96	-.10
Playing Quiet	1.40 (1.46)	0/5	.79	-.37
Motor	1.76 (1.71)	0/6	.50	-1.05
Talks	2.31 (1.85)	0/6	.39	-.99
Blurts	2.00 (1.74)	0/6	.47	-.93
Waiting Turn	1.63 (1.67)	0/6	.84	-.25
Interrupts	1.99 (1.81)	0/6	.56	-.74
EF Tasks				
NAVON Switch Cost (ms)	353.79 (302.78)	-457.40/1678.20	.62	2.04
WISC WMI (Standard Score)	99.21 (10.36)	68/129	-.11	.68
DM Tasks (Drift Rate)				
Choice Discrimination	2.53 (1.12)	.78/6.52	.98	.61
Mental Rotation	.81 (.29)	.24/1.76	.74	.45

H = Primarily hyperactive/impulsive subtype; I = Primarily inattentive subtype; C = Primarily combined subtype; ODD = Oppositional Defiant Disorder; CD = Conduct Disorder; GAD = Generalized Anxiety Disorder; MDD = Major Depressive Disorder; DD = Dysthymic Disorder; EF = Executive Functions; WMI = Working Memory Index; DM = Diffusion Model.

^aChildren meeting full diagnostic criteria based on structured diagnostic interview (DISC-IV; Shaffer et al., 1997) with primary caregiver and (for ADHD) standardized parent and teacher behavioral rating scales (BASC, Conners).

Table 2. Correlations of manifest variables.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
1 Mistakes	1																					
2 Sustains Attn	.81**	1																				
3 Listens	.70**	.79**	1																			
4 Follows	.74**	.75**	.73**	1																		
5 Disorganized	.80**	.85**	.76**	.83**	1																	
6 Avoids	.71**	.75**	.63**	.74**	.76**	1																
7 Loses	.73**	.70**	.65**	.73**	.79**	.64**	1															
8 Distracted	.78**	.74**	.70**	.77**	.80**	.70**	.70**	1														
9 Forgetful	.78**	.74**	.70**	.77**	.80**	.63**	.75**	.74**	1													
10 Fidgets	.58**	.71**	.63**	.61**	.63**	.54**	.53**	.73**	.58**	1												
11 Leaves Seat	.46**	.60**	.52**	.55**	.51**	.44**	.44**	.66**	.47**	.77**	1											
12 Runs/Climbs	.43**	.46**	.46**	.42**	.42**	.36**	.35**	.53**	.40**	.66**	.73**	1										
13 Quiet	.48**	.53**	.51**	.47**	.47**	.48**	.47**	.59**	.42**	.62**	.65**	.63**	1									
14 Motor	.50**	.58**	.52**	.46**	.49**	.36**	.46**	.59**	.46**	.72**	.75**	.73**	.66**	1								
15 Talks	.48**	.54**	.46**	.44**	.48**	.37**	.51**	.63**	.47**	.59**	.64**	.58**	.69**	.74**	1							
16 Blurts	.51**	.56**	.51**	.42**	.47**	.39**	.49**	.63**	.44**	.62**	.65**	.55**	.67**	.69**	.81**	1						
17 Waiting Turn	.47**	.56**	.52**	.47**	.48**	.39**	.44**	.62**	.42**	.66**	.73**	.64**	.66**	.73**	.72**	.77**	1					
18 Interrupts	.49**	.58**	.54**	.47**	.50**	.43**	.47**	.67**	.45**	.63**	.67**	.60**	.65**	.71**	.78**	.85**	.80**	1				
19 Choice Disc	-.24**	-.26**	-.31**	-.16*	-.24**	-.16*	-.22**	-.25**	-.16*	-.23**	-.23**	-.19**	-.21**	-.21**	-.17*	-.22**	-.23**	-.23**	1			
20 Mental Rot	-.20**	-.20**	-.22**	-.21**	-.19**	-.11	-.21**	-.17*	-.19**	-.12	-.19**	-.14*	-.16*	-.19**	-.19**	-.09	-.18*	-.11	.24**	1		
21 WMI	-.28**	-.29**	-.17*	-.22**	-.23**	-.28**	-.20**	-.23**	-.21**	-.18*	-.11	-.012	-.17*	-.12	-.17*	-.14*	-.12	-.21**	.13	.18*	1	
22 NAVON	.05	.11	.14	.14*	.17*	.10	.15*	.09	.11	.06	.05	.02	.12	-.04	.07	.07	.06	.06	-.09	-.16*	-.20**	1

Pearson correlations of manifest variables confirmed stronger relationships between the three impulsivity symptoms (i.e., blurting out answers, waiting turn, interrupting) than with hyperactivity symptoms. Given the conceptual overlap between impulsivity symptoms, correlations between their residuals were added to the model. Model 1 fit was good: χ^2 (114, $N = 201$) = 199.62, CFI = 0.965, TLI = 0.952, SRMR = 0.029, RMSEA = 0.061, 90% CI [0.047–0.075] (Table 3). ECV, PUC, ω , and H-index values are reported in Table 4 and all strongly supported the general ADHD factor as an interpretable representation of disorder symptomology. ω_s values indicated that neither specific factor captured sufficient reliable variance to warrant interpretation (both $\omega_s < .30$) (Gignac & Watkins, 2013). A hierarchical model in which the latent inattention and hyperactivity factors loaded onto a second-order ADHD factor was tested as an alternative model, but did not converge.

EF and drift rate as predictors of ADHD symptomology

In separate models, we first tested the degree to which standard EF metrics (Model 2a) and drift rate (Model 2b) independently predicted ADHD symptomology. All loadings for the latent EF (i.e., WISC WMI and NAVON switch cost) and drift rate factors (i.e., mental rotation and choice discrimination) were significant and in the expected directions (all $p \leq .005$), and both models fit the data well (see Table 3). In Model 2a, poor EF significantly predicted greater general ADHD symptoms (standardized estimate = -0.331 , $p = .019$), as did slow drift rate in Model 2b (standardized estimate = $-.487$, $p = .001$). There were no significant associations with the specific inattention or hyperactivity factors in either model (all $p \geq .081$).

In Model 3, both the latent drift rate and standard EF metric factors were included to examine whether their individual associations with ADHD symptomology survived after controlling for mutually predicted variance. Model 3 is depicted in Figure 1 and fit the data well: χ^2 (181, $N = 201$) = 216.54, CFI = 0.987, TLI = 0.984, SRMR = 0.031, RMSEA = 0.031, 90% CI [0.009–0.046].

Table 3. Model fit statistics.

Model	χ^2 (df)	CFI	TLI	RMSEA	SRMR
1	199.62 (114)*	.965	.952	.061 [.047, .075]	.029
2a	190.36 (146)*	.983	.978	.039 [.021, .054]	.030
2b	201.10 (146)*	.979	.973	.043 [.027, .057]	.030
3	216.54 (181)	.987	.984	.031 [.009, .046]	.031

For chi squares, $N = 201$. CFI = Comparative Fit Index; TLI = Tucker-Lewis Index; RMSEA = Root Mean Square Error of Approximation; SRMR = Standardized Root Mean Residual.

* $p < .01$

Table 4. Quality indices for bifactor model.

Factor	ECV	$\omega_{H/S}$	H
General	.70	.81	.96
Inattention	.16	.29	.75
Hyperactivity/Impulsivity	.14	.28	.72

Percent uncontaminated correlations = .53.

All factors significant at $p < .001$. ECV = Explained common variance; $\omega_{H/S}$ = Coefficient omega hierarchical/subscale; H = H-index.

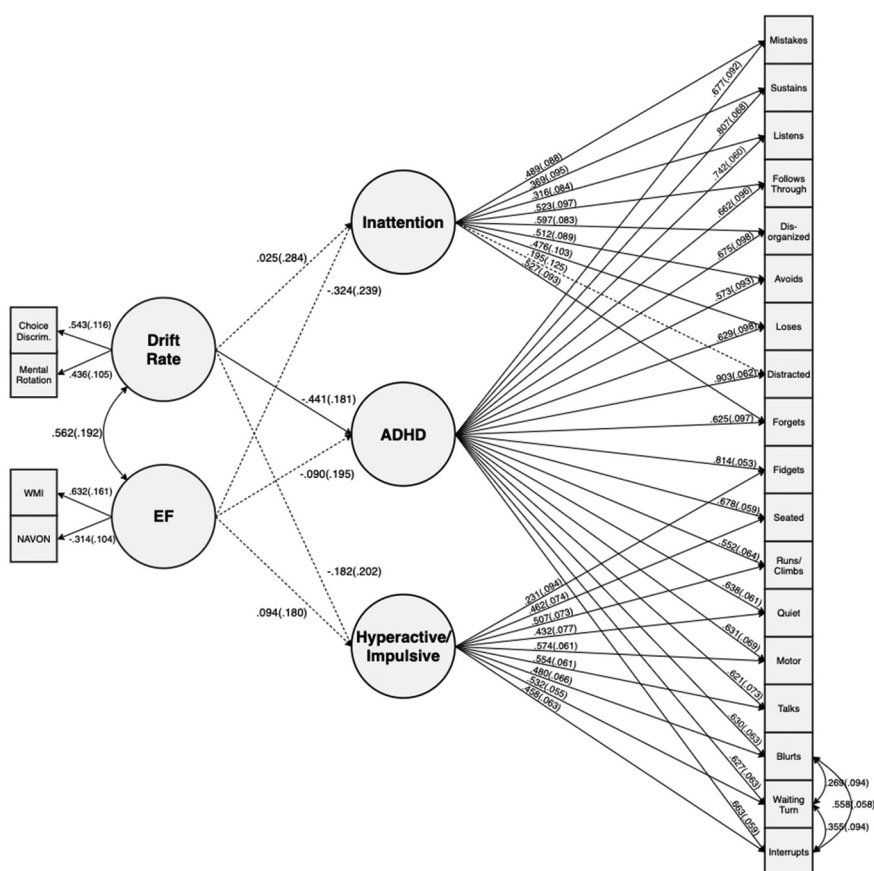


Figure 1. Model 3. Executive function and drift rate as latent predictors of the bifactor model of ADHD. Non-significant paths shown as dotted lines.

Slow drift rate continued to predict greater ADHD symptomology after accounting for the variance contributed by EF, with a standardized estimate of $-.441$ ($p = .030$). In contrast, the standardized estimate of $-.090$ for the association between EF and the general ADHD factor was not significant ($p = .642$). All associations with the specific inattention and hyperactive factors remained nonsignificant (all $p \geq .19$).

Discussion

“Executive dysfunction” is widely considered to be a candidate endophenotype and one etiological pathway through which ADHD manifests (Castellanos & Tannock, 2002; Rommelse et al., 2009; Sonuga-Barke, 2005; Sonuga-Barke et al., 2010). However, these theories have yet to reconcile two major concerns with the study of EF as a construct. First, measurement of EF relies on mean RT and accuracy, neither of which identifies clearly specified psychological processes responsible for poor performance among children with ADHD. Second, EF as a construct invokes a neurologically implausible “executive controller” or “homunculus” that governs complex cognition.

Recent work addressing these concerns has argued for the adoption of more neurologically grounded models of cognition to guide research. This movement proposes that regardless of whether the cognitive process of interest is “working memory,” “set shifting,” or “inhibition,” efficient action control is fundamentally determined by several steps including encoding, evidence accumulation, boundary separation, and response execution, all of which are well characterized by DM parameters (Ratcliff & McKoon, 2008; Verbruggen, McLaren et al., 2014). The present study used an SEM approach to determine whether slow evidence accumulation speed, indexed by the diffusion model parameter drift rate, better predicted informant ratings of ADHD symptomology than traditional EF metrics. Both poor EF and slow drift rate each predicted higher general ADHD symptomology. However, after accounting for shared variance, only drift rate continued to predict the general ADHD factor. That is, slow drift rate continued to capture additional variance in real-world ADHD symptoms over and above RT and accuracy on EF tasks.

These results strongly support slow drift rate as a discrete cognitive correlate of ADHD, adding to the growing number of studies documenting this impairment on a variety of EF tasks (Huang-Pollock, Ratcliff et al., 2017; Karalunas & Huang-Pollock, 2013; Metin et al., 2013; Salum, Sergeant et al., 2014; Salum, Sonuga-Barke et al., 2014; Weigard et al., 2019; Weigard & Huang-Pollock, 2017). Though the DM is not the only sophisticated approach used to examine cognitive dysfunction, it holds a key advantage in that the parameters are more resistant to subjective or inconsistent interpretation. Specifically, findings of ADHD-related deficits in drift rate have led to compelling and converging evidence of a task-nonspecific impairment in the ability to extract high quality decisional evidence from perceptual stimuli. In contrast, among the many interpretations of the ex-Gaussian parameter most discriminant of ADHD, tau (τ , representing the skew of the RT distribution), include attentional lapses (Buzy et al., 2009; Epstein et al., 2006; Leth-Steensen et al., 2000), the decision phase (Madden et al., 1999; Rotello & Zeng, 2008), WM/reasoning ability (Schmiedek et al., 2007), and central/attentional processing (Balota & Spieler, 1999; Spieler et al., 1996). Thus, the DM and the drift rate parameter offer a hitherto absent method of explaining cognitive deficits in ADHD in a clear, consistent, and mechanistically specific way.

Another advantage of the DM is that it incorporates both RT and accuracy into a single set of performance metrics, describing performance more completely than the standard practice of separately and/or selectively reporting speed and accuracy. Analyzing the entirety of the data enables the DM to quantify the independent influences of the multiple processes captured within summary metrics. This is especially important considering that these processes may influence performance in opposing directions. For example, slow drift rate will generally induce slower RTs. However, this effect can be offset by sufficiently fast nondecision time or narrow boundaries (Lerche & Voss, 2020). Multiple studies have documented this effect in ADHD samples (Feldman & Huang-Pollock, 2020; Karalunas & Huang-Pollock, 2013; Metin et al., 2013; Shapiro & Huang-Pollock, 2019; Weigard & Huang-Pollock, 2017), suggesting that a subset of children with ADHD who appear cognitively “unimpaired” based on standard metrics may still have marked impairments on a key cognitive component of decision-making. This enhanced specification of performance factors may explain why drift rate shows a stronger association with parent and teacher reported ADHD symptomology.

It of course bears mentioning that cognitive dysfunction is not unique to ADHD. Although the DM has been applied to ADHD more extensively than any other clinical population, atypicalities in drift rate as well as boundary separation have also been linked to anxiety, schizophrenia, and dysphoria (Moustafa et al., 2015; White et al., 2009, 2016). Considered alongside previous work from our group showing poor WM to predict externalizing symptomology above and beyond a general psychopathology factor (Huang-Pollock, Shapiro et al., 2017), the current findings also raise the question of whether drift rate may offer similarly improved transdiagnostic prediction of psychopathology relative to traditional measures of EF.

Importantly, the evidence accumulation process specified within the DM and broader cognitive literature has well-documented neurological support. Single-cell recordings in animals have shown gradually increasing rates of neuronal firing in the lateral intraparietal cortex (LIP) theorized to reflect the evidence accumulation process (Beck et al., 2008; Gold & Shadlen, 2007; Rodenacker et al., 2017). Similarly, EEG analyses using human participants have found a direct relationship between the drift rate parameter and activity in the parietal cortex (Nunez et al., 2017; Philiastides et al., 2006; Ratcliff et al., 2009, 2016). The promise of the DM as a bridge between neurology and cognition is exemplified by the “Neural Drift Diffusion Model” (NDDM; Turner et al., 2015). By integrating the DM and functional neuroimaging data at a single trial level, the NDDM showed an improved ability to predict behavior over either method alone. The continued development and implementation of such models will undoubtedly facilitate substantial advances in understanding cognitive function and dysfunction, including in clinical populations.

The study design afforded a number of strengths that lend further support to these results. First, the use of latent drift rate and EF factors comprised of multiple diffusion modeled and traditionally scored tasks (respectively) makes our findings more robust to concerns of task impurity and measurement error often cited in single-task analyses. Second, combining parent and teacher ratings created indicators that represented behavior in multiple environments (Martel et al., 2015). Latent factors comprised of averaged (equivalent to summed) parent and teacher symptom ratings have been shown to capture more variance and correlate more strongly with “gold standard” diagnoses (i.e., via clinical interview) relative to factors derived using the “or” (i.e., symptom present if endorsed by either rater) or the “and” algorithms (i.e., symptom present if endorsed by both raters) commonly used in research (Martel et al., 2015). With that said, the optimal means of combining informant ratings of ADHD symptoms remains an ongoing discussion in the literature. Third, the SEM framework allowed all regressions to run simultaneously while also accounting for shared variance between standard EF metrics and drift rate, making analyses less susceptible to Type I error than the use of separate regressions or bivariate correlations with factor scores.

Finally, consistent with prominent work supporting a bifactor model of psychopathology (Caspi et al., 2014; Lahey et al., 2012, 2017), a bifactor structure of ADHD has similarly been found to best model the disorder (Arias et al., 2018; Martel et al., 2010). The allocation of item variance primarily to the general vs. specific factors aligns with the widely accepted understanding of ADHD as a single clinical disorder. The inattention and hyperactivity domains themselves, while useful descriptors of clinical presentation at the time of assessment (Epstein & Loren, 2013; Willcutt et al., 2012), are

temporally unstable and do not differentially predict impairment (e.g., comorbidity, social skills, academic achievement; Lee et al., 2008; Owens et al., 2009). Indeed, more strongly emphasizing these points was what prompted the change in terminology from ADHD “subtypes” to “presentations” in the DSM-5. Thus, in the current work, modeling ADHD as a bifactor structure affords interpretive confidence that drift rate was predictive of the ADHD syndrome. The fact that neither cognitive construct predicted the specific factors over and above the general ADHD factor would ordinarily suggest the absence of any additional associations with the broader behavioral constructs of inattention or hyperactivity, though low ω_s values (both < 0.3) caution against making any conclusive or meaningful interpretations involving the specific factors (Gignac & Watkins, 2013).

The present study is of course not without limitations. First, the sample size was somewhat small for SEM and given the relative complexity of the final model. Bifactor models are also prone to overfitting data for various reasons (e.g., skewed data, overflexibility in allocating variance), which can lead to cross-study statistical and interpretive instability of the general factor (Bornova et al., 2020). However, our model was consistent with those produced in multiple large-sample studies, and with the empirical norm of all indicators loading moderately to strongly onto the general ADHD factor and weakly to moderately onto the specific inattention and hyperactivity factors (Arias et al., 2018; Caci et al., 2016; Willoughby et al., 2015). This reassures us that the general factor was indeed a valid index of the core ADHD construct. Second, it is possible that findings would vary had one or more of the many extant alternative tasks tapping other components of WM (e.g., dual processing, updating) or other commonly explored EFs (e.g., inhibition) been incorporated. With that said, the WMI and NAVON tasks are respectively among the most widely accepted indices of WM and set shifting, two prototypical EFs. Thus, while future studies are encouraged to replicate findings using alternative EF constructs and tasks, those used in the present study were well-qualified for inclusion in the common EF variable. Finally, given the study’s goal of viewing the cognitive correlates of ADHD through a more neurologically realistic lens, simultaneously collected EEG and/or functional neuroimaging data to complement cognitive data would be a valuable addition to future work.

Conclusion

These findings bear important implications for the field’s current conceptualization of EF as a central tenet of ADHD. The disappearance of an association between ADHD and the latent EF metric factor suggest that slow drift rate is both a more parsimonious and psychologically meaningful explanation for cognitive dysfunction in the disorder. This highlights the critical need for the field to revise the standard approach to cognitive assessment to one that emphasizes full data inclusion and extraction of dissociable, quantifiable, and well-specified cognitive mechanisms. The theoretical base and methodological sophistication of the DM make it one promising avenue to this end. Greater empirical emphasis on the drift rate parameter in particular will likely lend to neuropsychological models of ADHD that are both more realistic and informative for subsequent research.

Disclosure statement

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